Name : ________________ Class : _________ ID : _______________

1 What are the two major components of a composite laminate and state their individual functions in the laminate itself?

ANS:-
Fiber:- The fibers in a composite are the primary load-carrying element of the composite material. The composite material/laminate will be strong and stiff in the direction of the fibers.

Matrix:- The matrix supports the fibers and bonds them together in the composite material. It transfers any applied loads to the fibers, keeping the fibers in their position and chosen orientation. The matrix determines the composite environmental resistance like corrosion resistance and the maximum service temperature of a composite structure.

2a What are “tape” and “fabric”?

ANS:
Unidirectional: The fibers run in one direction and the strength and stiffness are in the fiber direction only. Prepreg tape or Tape is an example of a unidirectional ply orientation.

Bidirectional: The fibers run in two directions, typically 90° apart. These ply orientations have strength in both directions but not necessarily the same strength. A plain weave fabric or fabric is an example of a bidirectional ply orientation.

2b What does isotropic mean and how does composite laminate achieve quasi-isotropic?

ANS:
Isotropic means that the material has the same properties when tested in different directions.

Quasi-isotropic: The plies of a quasi-isotropic lay-up are stacked in a 0°, -45°, 45°, and 90° sequence or in a 0°, -60°, and +60° sequence. These types of ply orientations simulate the properties of an isotropic material such as aluminium or titanium.

2c Define symmetry and balanced for a composite laminate lay-up.

ANS:-
Symmetry of lay-up means that for each layer above the midplane of the laminate there must exist an identical layer (same thickness, material properties, and angular orientation) below the midplane.
Balance of a lay-up means that for every layer centered at some positive angle there must exist an identical layer oriented at a negative angle with the same thickness and material properties.

3a Currently, fiberglass and carbon composites are used more extensively on the modern aircraft. State the advantages and disadvantages between these two types of fibers

ANS:
Fiberglass:-

- Inherent electrical property
- Good chemical and galvanic corrosion resistance.
- Cheaper than carbon fibers

- Typically used for secondary structures on aircraft.
- Properties of fiberglass are less than carbon but lower cost than other carbon fiber.

Carbon / Graphite:-

- High strength compared to other metallic or composite materials.
- Superior fatigue property.
- High resistance to corrosion and creep.

- Galvanic corrosion with metallic fasteners and structures.
- Lower electric conductivity compared to aluminium (1000 times less).

3b What is Lightning Protection Fiber

ANS:

- Lightning Protection Fiber is not really a fiber, but a metallic mesh or foil, typically on top of the composite laminate to increase electrical conductivity
- Bronze and Copper used instead of Aluminium for carbon composite.

4a Why are the advantages and disadvantages of epoxy as a matrix materials?
ANS:

- Extended service temperature range
- Excellent mechanical properties: High strength and modulus
- Low levels of volatiles
- Excellent adhesion
- Low shrinkage (Little change in volume after cured)

- Good chemical resistance
- Ease of processing

- Brittleness
- Reduction of properties in the presence of moisture.

4b State the greatest advantage of the following plastic resin used to make aircraft composites?
- Bismaleimide (BMI) resins (A type of polyimide)
- Phenolic (Phenolformaldehyde) resins
- Polyetheretherketone (PEEK)

ANS:
Bismaleimide (BMI) resins (A type of polyimide):
- A thermosetting resin
- Are used when a higher service temperature is required, for example aero engines and high-temperature components.
- It require higher cure temperatures, typically 375 to 450°F and have epoxy-like processing characteristics yet have a higher temperature use limit.

Phenolic (Phenolformaldehyde) resins:
- A thermosetting resin
- Are used to manufacture composite for interior components because of their low smoke and flammability characteristics (Good Fire Smoke Toxicity (FST) properties).
- Less expensive alternative for lower temperature component.

Polyetheretherketone (PEEK):
- A thermoplastic resin
- An often used resin system for the aerospace industry to make structural parts to withstand impact
- It has superior toughness and good mechanical properties at elevated temperatures and after impact

5a What is pre-preg and how is pre-preg different from wet-up lay-up.

ANS:
Pre-preg consists of a combination of a matrix and fiber reinforcement. In other words, the pre-preg are either tape (unidirectional) or fabrics (bi-directional) pre-impregnated with the matrix/resin at factory side. Thus it must be stored in a freezer at a temperature below 0°F to retard the curing process (Stage B).

Wet lay-ups is when you impregnate the tape or fabric yourself with a matrix/resin of your choice.

The reason why pre-preg is often preferred is because it cures at a higher temperature, thus more strength. Also, being pre-impregnated with resin at the factory side, it reduces possibility of contamination as factory product abide to a strict condition and specialised equipment.

5b Define what is curing and the three types of curing stages?

ANS:
Curing is a term which refers to the toughening or hardening of a polymer material by cross-linking of polymer chains, brought about by electron beams, heat or chemical additives.

A Stage: The components of the resin (base material and hardener) have been mixed but the chemical reaction has not started.

The resin is in the A stage during a wet lay-up procedure.
B Stage: The components of the resin have been mixed and the chemical reaction has started. Material has thickened and is tacky.

Pre-preg materials are in the B stage. To prevent further curing, the resin is frozen. Any increase in temperature will start the cure cycle.
C Stage: The resin is fully cured.

6 What are the three types of adhesives and what are they typically used for?

ANS:
Film adhesive: Thin films supported on a release paper and stored under refrigerated conditions (-18°C, or 0°F).
Typically used as a high strength adhesive to bond honeycomb to facesheet, precured detail parts and pre-pregs to original structures.

Paste Adhesive: Two components systems requiring careful weighing and thorough mixing to ensure strength is not compromised.
Typically used in the repair of composite parts as filler materials, for bonding repair core sections in place, and for bonding repair patches.

Foaming adhesive: Consist of a thin unsupported epoxy film containing a blowing agent. During the rise to cure temperature, an inert gas is liberated causing an expansion or foaming action in the film.
Typically used as a lightweight splice material for splicing honeycomb core repair sections.

7 Describe how a sandwich structure is constructed.

ANS: A "sandwich" is constructed using at least two layers of high-strength material for facings (facesheet) and a lightweight, lower strength material for a core.

8 When does a sandwich panel is preferred over a laminate ?

ANS:If two layers of fiberglass laminates are placed on each side of a foam core, the stiffness (resistance to buckling) will be many times greater than when four fiberglass layers form a solid laminate.
The foam core, being lightweight, produces an almost negligible increase in weight. In other words, a sandwich panel is typically stiffer than a solid laminate itself.

9a What materials can be used for a core in a sandwich panel (composite)?

ANS:
It can be any materials but generally there are TWO types:-
Solid materials: Balsa wood, Plastics, Foams, Metals, Nomex (Paper impregnated in Phenolic resins),
Honeycomb materials: Can be made from same materials as solid materials but is in honeycomb form

9b For the core in a sandwich structure, it is typically honeycomb for aircraft panels. State THREE types of honeycomb cells and explain what it means when one honeycomb has cells smaller then another.

ANS:
- Hexagonal cells
- Overexpanded cells
- Flexicore

When the cells of a honeycomb core are smaller than another, it means that the density is higher and thus stronger and stiffer but also heavier.

10 Why are composite primary structures being made as laminate structure instead of sandwich structure?

ANS:-
Laminate structures are used for primary structures like wings and fuselage because:
- More damage tolerant.
- Easier to repair.
- Easier to manufacture with automated manufacturing techniques.
- Co-curing can be done with two or more laminated structures/components to eliminate secondary bonding processes or the use of fasteners.

OR
Sandwich structures has the following disadvantages:
- Low impact damage resistance.
- Prone to water intrusion.
- Difficult to implement bolted repair. Typically, the core has to be potted with resin before installation of fasteners.*
- Shapes/Contours of panels is dependent on the type and thickness of core used.*

11a What is the difference between delamination and debond/disbond for a composite damage.

ANS:
Delaminations form on the interface between the layers in the laminate. Debonds or disbonds can form along the bondline between two elements.

11b Why must water be removed from the panel once there is water ingression.

ANS:
Water create additional damage in repaired parts when cured above boiling point. Removal of water before repair is cured is recommended.

Freeze-Thaw Cycle: Water ingression will expand in volume as ice during flight and melt when on ground. This cycle repeats every flight and may lead to severe delamination or debond.

11c Why do composite structures need to be painted immediately after production from factory?

ANS:
Composite structures have to be protected by paint to prevent the effects of UV light degrading the plastic resins (matrix) used.

12a Explain what is audible sonic testing (coin tapping/tap test) and its limitations.

ANS:
This NDT consists of lightly tapping the surface of the part with a coin or other suitable object. Acoustic response of the damaged area is compared with that of a known good area. A dull sound is a good indication that some delamination or disbond exists, although a clear, sharp tapping sound does not necessarily ensure the absence of damage, especially in thick panels. Its limitation is that the acoustic response of a good part can also vary dramatically with changes in geometry so the tester needs to understand the component's contours well. Also this inspection loses its reliability for laminate with 4 plies or more. It is only good for detecting delaminations or disbond/debond.

12b Explain what is thermography in terms of non-destructive testing and its limitations.

ANS:
All thermographic techniques rely on differentials in thermal conductivity between normal, defect-free areas and those having a defect. Defect-free areas conduct heat more efficiently than areas with defects, the amount of heat that is either absorbed or reflected indicates the quality of the bond.

Its limitation is that the test subject must be subjected to a difference in temperature. At the same time, it is most effective for thin laminates or for defects near the surface. Also, It is most effective for detecting delaminations or disbond/debond or water ingression.

13 What is cosmetic repair and why is it carried out?

ANS: It is a repair which does not improve or affect the structural integrity of the component or aircraft. Instead, it is carried out to either protect the surface or to decorate or improve the aesthetic of the existing damage. It will not involve the use of reinforcement materials. For example, dents on passengers are common and are often good to be leave as it is provided the depth of dents is within limits. However, it might not be pleasing to passengers and thus a cosmetic repair is to be done.

14 State the typical steps for a composite laminate fabrication.

ANS:
1. Tape/Fabric plies layup: Plies lay-up is done per laminate map and a mould
2. Vacuum Bagging: Vaccum bagging is the most common method used to apply pressure during the curing of the composite.
3. Curing: Curing is a term which refers to the toughening or hardening of a polymer material by cross-linking of polymer chains, brought about by electron beams, heat or chemical additives. Typically, the higher the cure temperature, the stronger the bond is
4. Final inspection: Depending on the quality acceptance of the laminate, inspection can range from visual to NDT (typically ultra-sonic inspection)

15 When do you NOT need a bleeder ply during vacuum bagging?

The traditional way of bleed out using a vacuum bag technique is to use a perforated release film and a breather material / bleeder ply on top of the repair. Holes in the release film allow air to breathe and resin to bleed off over the repair area. If the lay-up is thick, the top plies tend to be resin starved and the bottom plies, resin rich.

However, some pre-pregs with 32 to 35 percent resin content are typically no-bleed systems as they contain exactly the amount of resin needed in the cured laminate. Bleed out of these pre-pregs will result in a resin-starved repair or part. A sheet of solid release film (no holes) will be placed on top of the pre-preg during the vacuum bagging to prevent bleed out instead.

16 What are the advantages and disadvantages of doing wet lay-up and pre-pregs for repair?

ANS:
<u>Wet lay-up</u>

Advantages: Wet lay-up cure at room temperature is easy to accomplish and the separate materials can be stored at room temperature for a long time.

Disadvantages: Room temperature wet lay-up will not restore the strength and durability of the original structure, which was cured at a much higher temperature in the first place.

<u>Pre-pregs</u>

Advantages: Pre-pregs cure at high temperature (typically around 120°C to 180°C), which are stronger. Typically, primary structure are made from pre-pregs and using pre-pregs for repair is good for restoring the strength.

Disadvantages: Pre-pregs requires freezer storage and hard to transport due to the requirement for low temperature.

17 What are the advantages and disadvantages of the following bonding processes?

- Co-curing
- Secondary bonding
- Co-bonding

ANS:
<u>Co-curing</u>
Is a process wherein two parts are simultaneously cured together.
Advantages are excellent fit between bonded components and guaranteed surface cleanliness.

<u>Co-bonding</u>
One of the detail parts is pre-cured, with the mating part being cured simultaneously with the adhesive.
Film adhesive is often used to improve peel strength.

<u>Secondary bonding</u>
Ultilises pre-cured composite detail parts and uses a layer of adhesive to bond two pre-cured composite part together.
Pre-cured parts usually have peel ply cured onto the secondary bonding surfaces.

18 Why is autoclave being used by MROs or OEMs to cure primary structures?

ANS:
Autoclaves are used in the curing process because they are able to be accurately controlled with temperature as high as 650°F and pressure up to 250 psi. During the curing process, the high pressure and temperature allows removal any trapped air or moisture from the composite material for proper curing resulting in a high-strength and durable structure. This also allows for

the curing of complex shapes and structures using proper moulds, which may not be possible with other curing methods.

19 What are the advantages and disadvantages between bolted repair and bonded repair for composite?

Bolted Repair	Bonded Repair
Repair typically used on composite thicker than 0.125” to ensure sufficient bearing area for load transfer.	Thick composite repair requires several curing sessions or the use of autoclave.
Faster processing time.	Slower processing time.
Repair may be Category B (Require inspections).	**Category A (Permanent Repair)**
Repair material not sensitive.	Repair material is time, temperature, and moisture sensitive
Heavier repair (additional weight)	**Lightweight.**
Hole drilling must be done with care and caution. Introduction of new holes reduce cross-sectional area.	**No protruding parts or additional fastener holes.**
Have to pot the core for sandwich structure before fasteners can be installed.	**More suitable for sandwich structure as there is no potential for moisture ingression and no bearing load on the thin face sheets.**
No risk of heat damage due to curing.	Risk of heat damage on original structure and porosity forming during the curing of repair plies

20 What are the steps taken for a typical bonded repair?

ANS:

1. Evaluate the damage: To determine extent of the damage using visual inspection and proper NDI inspection.
2. Remove the damaged area: Take care not to enlarge the repair area during damage removal.
3. Taper sand a uniform taper around repair area: To taper sand accordingly per the type of repair to be implemented.
4. Dry and clean the repair area: To ensure no moisture prior to the curing of the bonded repair. Cleanliness is important to ensure proper adhesion.
5. Replace the damaged core: To cut a replacement core to size and to be bonded to the repaired area prior to patch curing.
6. Prepare the repair plies and lay-up: To prepare the repair plies per manual requirement and lay the repair plies per original orientation. Typically, 1-2 extra plies are added. Additional fiberglass ply may be applied for smoothing purposes.

7. Vacuum bag the repair: To apply pressure to the repair during curing.
8. Cure the repair: Ensure proper cure cycle is applied.
9. Remove vacuum bag and inspect the repair: To ensure structural integrity of the repair, NDI may be required depending on repair procedure chosen and size of repair.

21a What are the three types of bonded patch repair for a laminate?

ANS:
Scarf repair: In this type of repair, the damaged area is removed and the remaining material is tapered to create a smooth transition from the undamaged area to the repaired area. The bonded patch is then bonded to the tapered surface using adhesive film.

Stepped repair: In this type of repair, the damaged area is removed and the damaged area is cut at a right angle to create a stepped surface. The bonded patch is then bonded to the stepped surface using adhesive film. This repair is restricted only to flat surfaces or panels as it is impossible to achieve the steps in contoured panels.

External patch repair: In this type of repair, repair plies are bonded directly over the damaged area using adhesive film. The repair plies are typically larger than the damaged area to ensure proper bonding and to provide additional strength to the repaired structure.

21b Describe what is a potting repair

ANS:
Potting repair is done by filling up the damaged area with potting compound, it may be mixed with chopped fibers or materials.

These repairs does not restore the full strength of the damaged area. And used as a filler for cosmetic repair to edges and skin panels or also used in sandwich honeycomb panels as hard points for bolts and screws.

21c Describe what is a resin injection repair

ANS:
Resin injection repair is done by drilling TWO holes at opposite ends of the damaged area within the composite structure. Resin is injected into one hole until it flows out from the other holes. Pressure and heat are then applied to ensure proper curing of the resin injected as a repair.

22 What are pre-cured composite repair patch or doubler?

Pre-cured patch are composite patches made in advance and then cut to shape to be used in bonded repair.
It can also used for bolted repair provided that the thickness and contour fits.

<u>Advantages</u>

- Can be made in advance in the autoclave
- Have the same mechanical properties as the original aircraft structure
- Easier to bond the patch to the aircraft.
- Pre-cured patch of the same material of the original structure is preferred as a repair doubler for a bolted repair. It will reduce internal stresses due to different in thermal expansion.

Disadvantages

- Difficulty of using pre-cured patch to fit to the contour of the damage area.

23 What are the differences between fasteners used for metal and composite structures?

ANS:

- For composite materials, fastener materials are limited to titanium alloy, monel or stainless steel alloy.
- The footprint diameter of nuts and collars used on composite laminates/panels are larger than those used on metal.

24 What are the things to take note when doing bolted repairs on composite compared to sheet-metal repair?

- Pre-cured patch of the same material of the original structure is preferred as a repair doubler for a bolted repair. It will reduce internal stresses due to different in thermal expansion.
- Repair materials can be any of the three materials : Titanium alloy, Monel or Stainless Steel alloy
- Riveting in composite is restricted due to bucking of rivets can cause delamination. Solid rivets may be used in composites but washers are required on the driven head side to prevent delaminating of the laminate.
- Carbon fibers composites are cathodic. Care must be exercise to avoid contact with materials such as aluminum or cadmium. If contact is unavoidable, ensure there is a fiberglass ply separating direct contact between the metal and the carbon composite.
- Higher drill speeds (to avoid heat generation) and lower feeds (to avoid splintering at drill entrance and exit) are required to drill precision holes.
- During trimming or drilling of the composites, it should be well supported and firmly backed up with wood or an aluminium plate to prevent backside breakout.
- Diamond-tipped or carbide cutting and drilling tools are preferred and there are specialised drill bit or cutting bit meant for composite.
- Edge margin for composite material is to be at least 3D unlike 2D for sheet metal.